

Technical Report

EuMINE DataBridge Hackathon 2026

Team Name: **TakeMe2Romania**

Institutions: 1. University of Sarajevo, Bosnia and Herzegovina

2. NOMATEN CoE: National Centre for Nuclear Research, Poland

Contact emails: kbecarevic1@etf.unsa.ba, aspahic7@etf.unsa.ba, amustafic5@etf.unsa.ba

Date submitted: 22 June 2026

Abstract

Our approach extends the standard MAGPIE + Random Forest baseline by combining composition-based MAGPIE descriptors with additional structural, thermodynamic, and metadata features obtained from the integrated Materials Project and JARVIS datasets. While the baseline relies primarily on elemental composition statistics, the proposed model incorporates information such as crystal system, space group, density, and energy above hull, providing a richer description of material properties. The predictive model consists of a multi-output Random Forest ensemble with 200 decision trees and a maximum depth of 30, trained on a merged dataset containing approximately 165,000 materials. This strategy improves the model's ability to capture structure-property relationships while retaining the robustness and interpretability of tree-based methods. On the validation set, the model achieved a mean absolute error (MAE) of 0.3466 eV for band gap prediction and 0.1170 eV/atom for formation energy prediction.

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1 Model Architecture

1.1 Overview

We employed a composition-based machine learning ensemble model rather than a graph neural network, equivariant model, or transformer. The model uses MAGPIE descriptors extracted from the chemical composition of each material as input features. Specifically, crystal structures from Materials Project and JARVIS were converted into composition vectors using the MAGPIE featurization scheme implemented in Matminer, which computes statistical summaries (mean, maximum, minimum, range, standard deviation, etc.) of elemental properties such as atomic number, electronegativity, atomic radius, melting temperature, and valence electron counts. These descriptors provide a fixed-length numerical representation of each material without requiring atomic coordinates, bonding information, or crystal graph construction.

As the predictive model, we used a Random Forest ensemble regressor wrapped in a MultiOutputRegressor framework to simultaneously predict multiple material properties. The Random Forest consisted of 200 decision trees with a maximum tree depth of 30, enabling the model to capture complex non-linear relationships between composition-derived features and target properties while maintaining robustness against overfitting. Since Random Forests are tree-based ensembles, the model contains no neural network layers, activation functions, message-passing operations, or learned embeddings. Likewise, there is no graph aggregation or readout stage.

1.2 Input Representation

Crystal structures were represented using composition-based MAGPIE descriptors generated with Matminer. Each material’s chemical formula was converted into a fixed-length feature vector by computing statistical summaries (mean, minimum, maximum, range, standard deviation, etc.) of elemental properties such as atomic number, electronegativity, atomic radius, and valence electron counts. The final representation consisted of 134 numerical features per material. No structural information (atomic coordinates, lattice parameters, bonds, or crystal graphs) was used; therefore, the model relies solely on composition-based descriptors rather than graph representations, point clouds, voxel grids, or pre-trained crystal embeddings.

In addition to the 134 MAGPIE composition-based descriptors generated using Matminer, several auxiliary compositional, structural, thermodynamic, and metadata attributes were incorporated into the feature set. These included the original and normalized chemical formulas (formula, formula_norm), thermodynamic properties such as energy_above_hull_eV/atom, structural information including crystal_system and spacegroup_number, material density (density_g/cm³), and dataset-specific attributes such as group, source_db, and functional. The dataset also contained unique identifiers (jid, material_id), data integration metadata (merge_date), filtering parameters (density_threshold, exfol_threshold_meV), and a composition object used for MAGPIE feature generation. The target properties were bandgap_eV and formation_energy_eV_per_atom, which were predicted by the model and therefore were not used as input features. By combining MAGPIE descriptors with additional structural and thermodynamic attributes, the model leveraged both elemental composition information and higher-level material characteristics available from the integrated Materials Project and JARVIS datasets.

1.3 Architecture Details

The implemented model is a composition-based Random Forest ensemble using MAGPIE descriptors as input features. It is not a neural network, graph neural network, transformer, or equivariant model; therefore, concepts such as layers, activation functions, and message-passing readout are not applicable.

The base model is a `RandomForestRegressor` with `n_estimators=200`, `max_depth=30`, `random_state=42`, and `n_jobs=-1`. Thus, the model consists of 300 decision trees. The exact number of parameters is not fixed in advance, because it depends on the number of tree nodes created during training. Unlike neural networks, the model does not learn weights and biases through backpropagation.

Each tree makes predictions by recursively splitting the MAGPIE feature space using threshold-based decision rules. The aggregation/readout function is the averaging of predictions across all trees in the Random Forest.

Because two properties are predicted, the model uses `MultiOutputRegressor`. This trains one independent Random Forest regressor for each output property: one for `bandgap_eV` and one for `formation_energy_eV_per_atom`. Therefore, the implementation is multi-output at the interface level, but internally it uses one separate output model per property.

Random Forest was selected because it provides a strong and computationally efficient baseline for tabular materials descriptors. The model trains rapidly, requires minimal hyperparameter tuning, and performs well on engineered composition-based features such as MAGPIE descriptors. While deep learning approaches, particularly graph neural networks, can achieve higher accuracy by explicitly modeling crystal structures and atomic interactions, they require substantially greater computational resources, larger training datasets, and more extensive hyperparameter optimization. Consequently, Random Forest offers an attractive trade-off between predictive performance, interpretability, and computational cost for composition-based materials property prediction.

2 Training Procedure

2.1 Data Split Used

The model was trained using a large merged dataset constructed from the Materials Project and JARVIS databases. After harmonization and preprocessing, the combined dataset contained approximately 165,000 materials with associated compositional, structural, and thermodynamic properties. An internal 80/20 split was applied, resulting in 132,426 training samples and 33,106 validation samples.

The notebook implementation does not provide evidence that the official benchmark split of 700 training, 150 validation, and 150 test samples was used during model training. Instead, the Random Forest model was developed and validated on the substantially larger merged Materials Project–JARVIS dataset. Additional data sources beyond the benchmark dataset were therefore incorporated, specifically the Materials Project and JARVIS repositories, to increase the diversity and coverage of materials available for training.

The merged dataset included composition information used to generate MAGPIE descriptors, as well as auxiliary structural and thermodynamic attributes such as crystal system, space group, density, and energy above hull. These additional data sources significantly expanded the training set compared to the benchmark dataset and enabled the construction of a more robust composition-based baseline model.

2.2 Loss Function

The Random Forest model uses mean squared error (MSE) as the split criterion during tree construction. Multi-property prediction is implemented through `MultiOutputRegressor`, which trains one independent Random Forest for each target property. Therefore, no explicit combined multi-task loss is used; each property is optimized separately. Model performance is evaluated using MAE, RMSE, and R^2 metrics.

2.3 Optimiser and Hyperparameters

Since the proposed method is based on a Random Forest ensemble rather than a neural network, hyperparameters related to gradient-based optimization are not applicable. In particular, no optimizer, learning rate scheduler, batch size, weight decay, or training epochs are used, as the model is trained by constructing decision trees rather than by iterative backpropagation.

2.4 Hardware and Runtime

Model training was performed using the scikit-learn implementation of Random Forests. The model was executed on a CPU-based environment and utilized all available processor cores through the `n_jobs=-1` setting. Although the notebook imports GPU-enabled libraries, the Random Forest training itself was performed on the CPU, as scikit-learn’s Random Forest implementation does not require GPU acceleration.

3 Results

Table 1: Validation-set performance vs. baseline.

Model	MAE E_f (eV/atom) (speedup)	MAE E_g (eV) (speedup)
MAGPIE + RandomForest (baseline)	0.2657	0.5021
EXTENDED MAGPIE + RandomForest	0.1170 (2.27x)	0.3466 (1.44x)

4 Reproducibility

4.1 Code Repository

The link to our code is provided on Github: https://github.com/becarevicKemal22/eumine_hackathon_2026.

4.2 Dependencies

The implementation was developed in Python and relied on several widely used libraries for materials informatics and machine learning. Material compositions were represented using the `pymatgen` library, while MAGPIE descriptors were generated using the `matminer` feature extraction framework. Data preprocessing and manipulation were performed using `pandas` and `NumPy`. The predictive model was implemented using `scikit-learn`, specifically the `RandomForestRegressor` and `MultiOutputRegressor` classes. Additional utilities from the scientific Python ecosystem were used for data analysis and evaluation.

4.3 Reproducing the Results

The steps that can be taken to download and run our model on the test structures are given in the listing below.

```
1 # 1. Navigate to our folder with scripts
2 cd submissions/TakeMe2Romania
3 # 2. Install required dependencies
4 pip install -r requirements.txt
```

At this point, it is necessary to copy the `test_input_structures.zip` that is provided by EuMINE into this `submissions/TakeMe2Romania` folder and unzip it so that the path of the `.cif` files is `submissions/-TakeMe2Romania/test_input_structures/`.

```
1 # 3. Run the prediction script
2 # This script will automatically download the pre-trained model,
3 # read the test CIF files from the 'test_input_structures' folder,
4 # compute descriptors, and generate the 'predictions_test.json' file.
5 python main.py
```

5 Discussion

5.1 Key Findings

We successfully constructed a unified dataset containing 165,533 materials through a structure-aware “smart merge” of the Materials Project and JARVIS-DFT databases. By classifying materials using properties such as exfoliation energy, density, and crystal symmetry, we assigned physically appropriate labels, retaining JARVIS OptB88vdW values for layered and two-dimensional materials (Group A) while prioritizing Materials Project PBE values for three-dimensional bulk solids (Group B). Using an internal 80/20 train/validation split on this expanded dataset (132,426 training and 33,106 validation samples) rather than the official benchmark split of 700/150/150 samples, we trained a multi-output Random Forest ensemble that extended the standard composition-based MAGPIE baseline by incorporating additional structural and thermodynamic descriptors. The resulting model demonstrated substantial performance improvements on the validation set, achieving a mean absolute error (MAE) of 0.3466 eV for band gap prediction (a $1.44\times$ improvement over the baseline) and 0.1170 eV/atom for formation energy prediction (a $2.27\times$ improvement). However, utilizing these cross-source labels introduced distinct challenges: because Group A carries OptB88vdW labels and Group B carries PBE values, the final dataset is not computed at a single, uniform level of theory. Our model implicitly learns a blend of two separate functionals, potentially introducing systematic errors at the Group A/B decision boundary that cannot be empirically quantified or corrected because our priority rules forced the two source subsets to be completely disjoint. Furthermore, this integration created a severe feature asymmetry, as exfoliation energy is a JARVIS-specific property completely unavailable for the 134,100 MP-origin rows, restricting its downstream use as a model feature exclusively to the JARVIS-origin subset. Ultimately, while these constraints highlight the limitations of mixed-functional datasets, our results emphasize the immense value of integrating

physically informed data harmonization with enriched feature engineering to improve materials property prediction while maintaining the efficiency and interpretability of tree-based machine learning models.

5.2 Limitations

The model may not perform well for materials that are very different from those present in the training data. Since it learns patterns from the Materials Project and JARVIS databases, predictions for rare element combinations or novel material classes may be less reliable.

The model is trained using properties obtained from density functional theory (DFT) calculations. Therefore, any systematic errors or approximations present in the DFT data may also be reflected in the model predictions. As a result, the predicted values may not always match experimental measurements.

Finally, the model relies on MAGPIE descriptors and other manually engineered features. It does not directly learn from atomic coordinates or crystal structures. Consequently, it may miss complex structure-property relationships that can be captured by modern graph neural network models.

5.3 Future Work

Given additional time, data, and computational resources, several improvements could be explored. First, the current Random Forest model could be replaced or complemented by structure-aware graph neural networks such as ALIGNN, M3GNet, or CHGNet. These models directly utilize atomic coordinates, bonding information, and local atomic environments, allowing them to capture structure-property relationships that are not accessible through composition-based descriptors alone.

References

1. Jain et al., *APL Mater.* **1**, 011002 (2013) — Materials Project.
2. Choudhary et al., *npj Comput. Mater.* **6**, 173 (2020) — JARVIS-DFT.
3. Ward, L., Agrawal, A., Choudhary, A. et al. A general-purpose machine learning framework for predicting properties of inorganic materials. *npj Comput Mater* 2, 16028 (2016). - MAGPIE